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A Study of Model-Based Protective Fast-Charging and Associated Degradation in Commercial Smartphone Cells: Insights on Cathode Degradation as a Result of Lithium Depositions on the Anode

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The ever expanding mobile consumer electronic market has accelerated the need for safe and efficient fast-charging approaches that improve the overall speed of battery charging without hastened deterioration of the battery performance. Herein, the impact of a resource inexpensive, physics-based, electrochemically optimized fast-charging algorithm (charging time < 2 h) for mobile devices is investigated. A critical difference in the amount and morphology of lithium deposits on the anode for cells fast-charged without an optimized algorithm is observed and found to be the main cause of capacity decay. An in-depth study of the $LiCoO_2$ cathode regions opposite to pronounced lithium deposits on the anode reveals a "mirroring" phenomenon, i.e., a frozen monoclinic phase, and inactivity to relithiation. In operando hard X-ray absorption spectroscopy reveals that degraded spots on harvested cathodes seem to be activated again and participate in the intercalation process when lithiated at low rates from lithium foil counter electrodes. On the other hand, tests at higher C-rates, closer to the actual fast-charging rate, reveal only negligible oxidation state changes and therefore poor performance.

1. Introduction

Since the commercialization of lithium-ion batteries (LIBs) in 1991, lithium cobalt oxide (Li_xCoO_2, LCO) is still the widest employed cathode material in batteries for mobile consumer electronics.[1,2] The success of LCO in portable electronics is mainly due to the steep material development progress in the last 20 years and the resulting long cycle life (>1000 cycles),[3,4] which is especially crucial for devices charged at least one to two times a day.[5–7] However, despite its high theoretical capacity of 274 mAh g^{-1}, LCO only delivers a useable specific capacity of about 140 mAh g^{-1} at an upper cutoff voltage of 4.2 V (vs Li/Li^+).[8–10] Hence, charging to higher cell cutoff potentials of 4.2–4.5 V has been a way to achieve higher specific energy.[11] However, at potentials above 4.5 V ($x = 0.28$), LCO shows a fast capacity fade, mainly due to instability of the layered structure and deleterious phase transition.[11,12] Cobalt dissolution,[9] oxygen loss,[13] and side reactions with the electrolyte[9] have also been shown to decrease lifetimes when cells are repeatedly charged to high potentials. Mitigating the issues with LCO at high potentials has attracted a lot of research interest and helped to improve the stability of LCO using surface modifications, including coatings,[14–18] and doping with a variety of elemental compositions.[18–20] Today, these efforts have led to LCO having the largest market share in terms of battery use for mobile consumer applications.[2,4] Besides offering high cycling stability for mobile consumer applications, another

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DOI: 10.1002/aenm.202003019

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requirement for LIBs utilized in smartphones and other portable electronic devices is the capability of fast-charging, without compromising cycle life. While modifications to the cathode material and cell design can improve the fast-charging capability,[21] the most efficient utilization of LIBs can be provided by Battery Management Systems (BMS) based on electrochemical models reflecting physical states inside the battery.[22–25]

BMS utilize a mathematical model of a battery to control its operation and typically refer to a hardware and software-based solution, where the software usually handles tasks such as the estimation of the state-of-charge (SOC) and state-of-health (SOH), charging algorithm, and establishing safe operating constraints. Models for BMS can range from empirical equivalent circuit based models to physics-based electrochemical models.[26] Empirical models have milder memory and computational requirements than physics-based models but do not provide insights into physical internal states of the battery that determine its performance and aging characteristics. Memory and computational capability of the hardware for the target applications typically determine the type of models used in the BMS. We discuss two potential barriers to the introduction of physics-based electrochemical model BMS in the area of consumer electronics applications for determining online fast-charging strategies and how they can be overcome.

1) Algorithms for parameter estimation for the electrochemical model: The performance of BMS in terms of accurate SOC/SOH estimation, power prediction, and online strategies for fast-charging is directly related to the quality of the initial model and the adaptation of model parameters as the battery ages. However, validating results of algorithms that estimate electrochemical model parameters is a time consuming process that requires specific experimental expertise.
2) Resource constraints: Algorithms for state and parameter estimation based on electrochemical models can be resource (power) intensive for real-time implementation in consumer electronics. Such models typically involve coupled nonlinear Partial Differential Equations that are solved by model order reduction techniques.[27] Despite having enough computation power, most consumer electronics still run on less resource intensive empirical models in order to preserve power for user applications.

2. Standalone Electrochemical Model

In order to get the full benefit of using physics-based electrochemical model BMS for fast-charging strategies, it is imperative to adapt the model parameters as the battery ages. The development, implementation, and validation of such algorithms are challenging. Generally, given a fully parameterized physics-based electrochemical model, i.e., containing all relevant electrolyte transport properties and electrode parameters, for a specific battery at a given age, an optimal charging profile that maximizes input charge current to the battery to reduce charging time can be derived. Such a charging profile is constrained to current, voltage, and temperature limits specified by the cell manufacturer while minimizing side reactions and other degradation mechanisms that could potentially age the battery faster such as lithium plating, electrolyte decomposition, and active material dissolution. However, such an approach is resource intensive. As an alternative we propose a standalone solution. The proposed approach maintains the same benefits as a physics-based electrochemical model for fast-charging applications, while utilizing simpler building blocks for charging profile generation and adaption rules that guarantee safe and efficient operation of the battery during charging, throughout the lifetime of the battery.[28] We use the following approach:

1) Given a specific battery, we parameterize electrochemical models at beginning-of-life (BoL), end-of-life (EoL), and at different stages in-between, through accelerated aging tests under operating conditions of interest.
2) Subsequently, for each aged state we optimize the charging strategies, i.e., minimize charging time, while reducing degradation at that age.
3) The developed charging strategies at each stage in the battery life are then approximated using simple building blocks such as constant current (CC), constant voltage (CV), and current ramps.
4) In a final step, adaptation rules and switching criteria are then developed to identify charging strategies at any age of the battery. Here, switching criteria refer to conditions based on SOC and SOH and are meant to trigger changes between different building blocks such as between CC to CV or CC to ramp current to CV and so forth.[29]

With this approach, we utilize the physical insights from electrochemical models, but use a simpler approach for real time implementation where only the building blocks and adaptation rules are required in addition to the estimate of the battery SOH for optimized battery charging.

The implementation of such a standalone electrochemical model-based charging strategy is demonstrated for the LG-G5 smartphone battery on an external battery cycler. We compare the impact of an Original Equipment Manufacturer (OEM)-specified fast-charging protocol and an electrochemical model-based fast-charging protocol on the capacity fade, swelling, and structural integrity of the active materials in anode and cathode. Using a complementary set of experimental characterization methods ranging from synchrotron in operando to lab-scale techniques, we show not only the functionality of an electrochemical model-based fast-charging strategy in consumer electronics but also its ability to suppress lithium plating. Finally, to the best of our knowledge, for the first time, we show how lithium depositions on the anode lead to inactivated cathode regions with a lower lithiation grade.

3. Performance Comparison of Cycling Protocols

The charging protocols considered include the OEM-specified fast-charging protocol (hereafter OEM_{cyc}) and a fast-charging protocol derived using a physics-based model (hereafter MOD_{cyc}) of the LG-G5 (BL-42D1F) LIB (for cell specifications see the Experimental Section); it is important to note here that "fast-charging" refers to a charging time of < 2 h. Details of the

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approximation procedure for the MOD_{cyc} fast-charging algorithm are disclosed in a US filed patent application.[30] Here we give a brief overview of the main differences between the OEM_{cyc} and MOD_{cyc} charge strategy (also see the Experimental Section). The OEM_{cyc} profile consists of a combination of CC and CV steps. The switch between CC steps and from CC to CV occurs after certain voltage thresholds are exceeded (top panel of **Figure 1**a). The OEM_{cyc} charging strategy is based exclusively on voltage and current limits and does not take into account the mechanisms that are responsible for battery degradation. For example, lithium plating is thermodynamically favored during charging if the anode surface overpotential becomes negative (with respect to Li/Li^+).[31] While it is feasible to ensure that lithium plating is not triggered at BoL by carefully choosing voltage and current limits for charging, these limits may become unreliable as the battery ages. Hence, charge profile adaptation and/or update of the BMS parameters over lifetime is imperative. Through carefully derating the current during battery charging at certain SOCs and adapting the strategy as the battery ages, the MOD_{cyc} approach prevents the battery from entering regimes that cause accelerated degradation of the battery capacity, while still staying within the current, voltage, temperature, and close to the charge time specifications of the OEM_{cyc} protocol. The difference between MOD_{cyc} and OEM_{cyc} protocol becomes clear from the current versus SOC plot at BoL and EoL for both protocols (top panels in Figure 1a,b). Although both protocols apply the same max. current at beginning of a charge cycle (3.105 A), the MOD_{cyc} protocol starts the current deration fairly early at around 0.2 SOC, while the OEM_{cyc} algorithm's hard cutoff is always set to be at a cell voltage of 4.3 V; this way higher currents are sustained for longer times and at higher SOCs for the OEM_{cyc} protocol, which can potentially cause degradation. At BoL the cutoff criterion of the OEM_{cyc} protocol is reached after ≈30 min, however, is reduced with prolonged cell lifetime to less than 10 min at EoL, which is mainly due to an impedance rise in the cell. The benefit of the MOD_{cyc} algorithm is also reflected when comparing voltage versus SOC for BoL and EoL and retained capacities. The dynamic voltage during charging with the OEM_{cyc} protocol is, bar none, always higher when comparing it to the dynamic voltage of cells cycled with the MOD_{cyc} protocol, which again could lead to more degradation. One representative example of the capacity retention for one pair of cells is shown in Figure 1c (see Figure S2, Supporting Information, for additional cycling data). In this example, the cell after MOD_{cyc} protocol (blue) retains 93.4% of its initial capacity after 115 cycles, while the cell after OEM_{cyc} protocol (red) only retains 84.6% of its initial capacity; cells cycled with both protocols are always evaluated after the same number of cycles. The MOD_{cyc} algorithm is designed such that it achieves comparable charging times to the OEM_{cyc} protocol while extending the cycle life; Figure 1d illustrates how the charging time for the MOD_{cyc} over the lifetime of the battery does not change significantly, whereas the charging with the OEM_{cyc} algorithm is, for roughly the first 50 cycles, around 5 min (≈4%) faster, but then starts to increase significantly. The initial decrease in charging time can be explained by the loss of capacity within the first 50 cycles for cells cycled with the OEM_{cyc} protocol and an associated increase in cell impedance. To elucidate causes of performance loss, we derived the dQ/dV versus cell voltage plots[32] for the slow $C/25$ discharge at 50, 100, and 115 cycles, as shown in Figure 1e–g. For both protocols, capacity loss is manifested as an overall decrease in area under the curve. With increased cycle number, intensities as well as areas of the peaks are decreasing, which is especially true for cells cycled with the OEM_{cyc} protocol, indicating a hastened capacity loss relative to the MOD_{cyc} protocol. The peak shift to higher voltages implies greater polarization, i.e., impedance rise (Figure 1g). This shift is mainly visible for the cell cycled with the OEM_{cyc} protocol and is less pronounced for the MOD_{cyc} protocol.

Besides preserving capacity during fast-charging, another important factor in cells for portable electronic devices is battery swelling due to the small form factor constraints and its minimization. The swelling, measured at 100% SOC, of cells charged via the MOD_{cyc} algorithm is also lower than that of the OEM_{cyc} cells (see Figure S3, Supporting Information), which is consistent with mitigation of lithium plating and/or accumulation of other solid side-reaction products. After 50 cycles, cells cycled with the MOD_{cyc} protocol only increased by 0.04–0.12 mm (1–3%), while cells cycled with the OEM_{cyc} protocol swelled by 0.18–0.45 mm (4.3–10.9%). Further cycling with the MOD_{cyc} protocol leads to a maximum swelling of 0.26–0.31 mm (6.3–7.5%, after 100 cycles), which does not further increase, even when cycled past 160 cycles. In contrast, cells cycled with the OEM_{cyc} protocol show swelling up to 0.70–0.75 mm after 120 cycles (16.9–18.2%, EoL); it has to be noted that the maximum tolerable swelling in modern smartphone batteries is between 7% and 9%. This significant difference is most likely due to increases in side reactions and possible gas evolution, for the cells cycled with the OEM_{cyc} protocol. Minimal swelling of the cells cycled with the MOD_{cyc} protocol gives an idea on how lithium plating,[33–35] depositions from side reactions, and possible solid electrolyte interface (SEI) growth, all contributing to cell swelling, can be minimized by an optimized fast-charging protocol.

4. Cycling Protocol-Induced Differences on the Anode

A schematic overview of the protocol-induced differences on the anode is visualized in **Figure 2**a. In general, a multitude of factors (low temperature cycling, high rate cycling, overcharging, and/or nonuniform charging) can affect the anode kinetics and the lithium ion diffusion rate and hence the tendency to deposit metallic lithium on the anode.[31,36–38] To confirm the presence of and classify lithium depositions on the anode, a visual inspection of anodes extracted from cells cycled with the OEM_{cyc} and MOD_{cyc} protocols was performed in the glove box (Figure 2b). To describe the macroscopic morphology of deposited lithium, we adopt terminology introduced by Waldmann et al.[39] Three common ways in which lithium can be deposited on the anode are, as follows: homogenous, covering the entire electrode surface, local, where hotspots of lithium can be found on the electrode, and marginal, where mainly edges show concentrated lithium deposits.[39] The term "lithium plating" is used to depict homogeneous lithium that evenly covers the electrode surface and "lithium deposition" may be used

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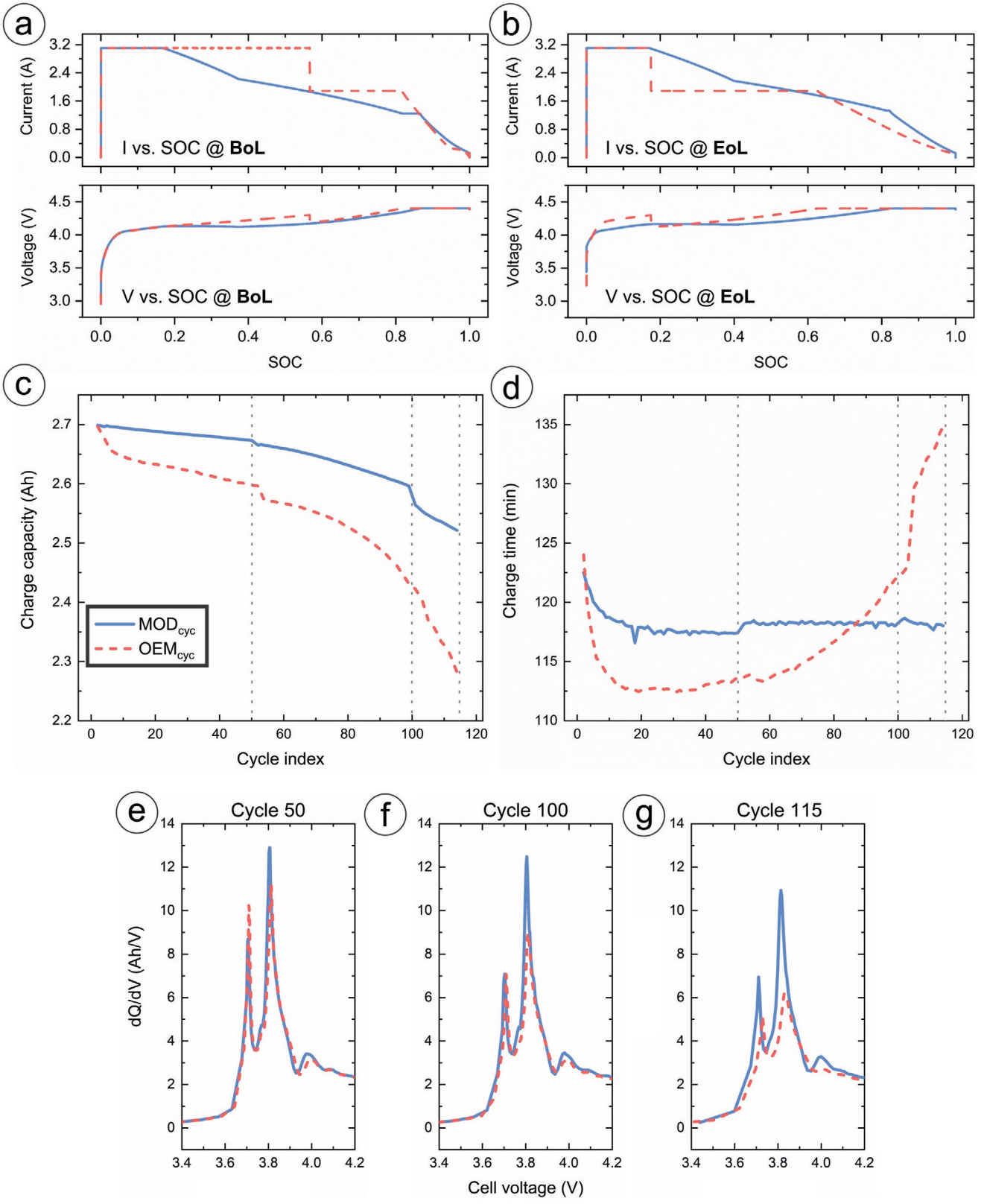


Figure 1. Electrochemical performance metrics for cells cycled with the MOD$_{cyc}$ protocol (blue) and with the OEM$_{cyc}$ protocol (red). a,b) Applied current versus SOC and dynamic cell voltage versus SOC for BoL and EoL, respectively (see Figure S1a,b, Supporting Information, for the respective current/voltage versus time plots). c,d) Capacity retention and charging times as a function of cycle index, respectively. e–g) dQ/dV versus V plots for MOD$_{cyc}$ and OEM$_{cyc}$ cells after 50, 100, and 115 cycles, as indicated by the dashed gray lines in (c) and (d).

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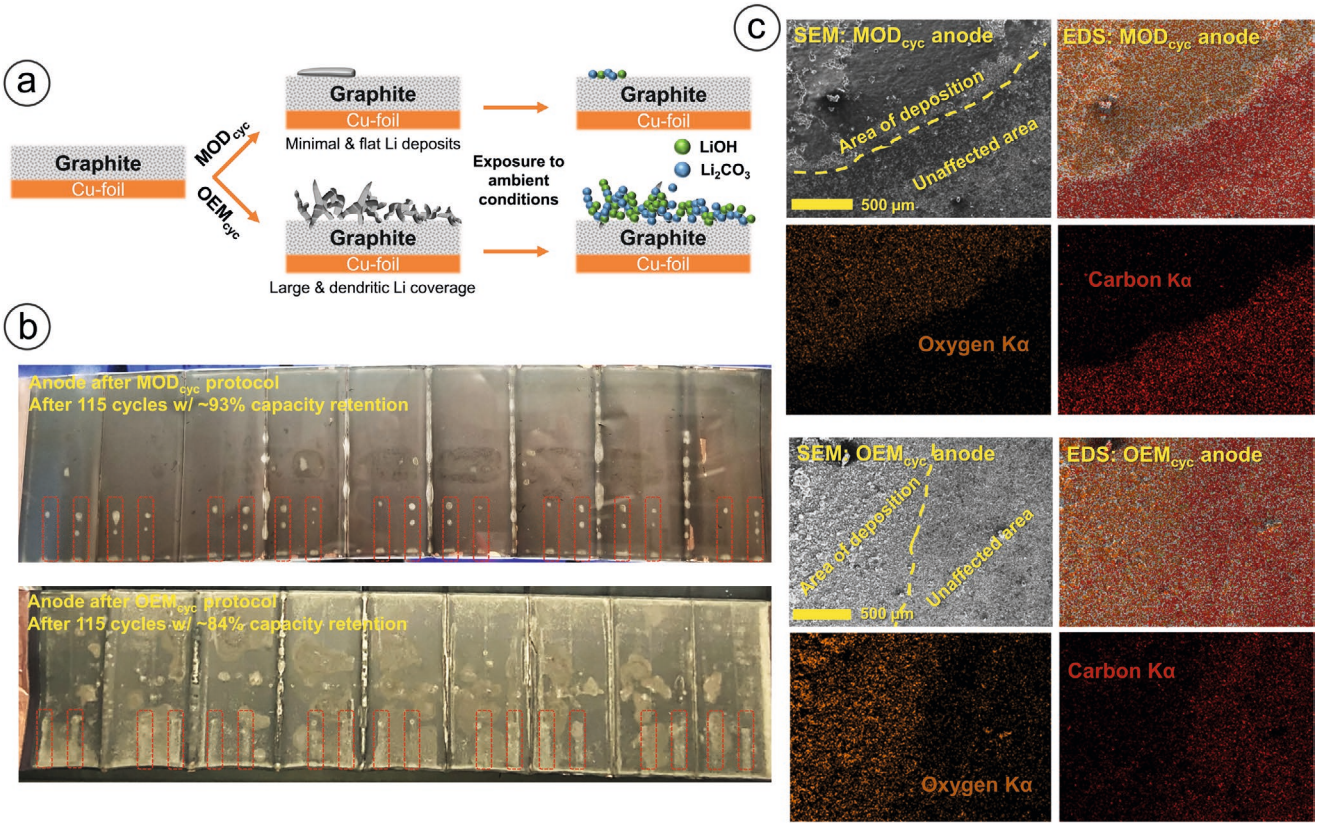


Figure 2. a) Schematic overview of the amount and morphological difference of deposited lithium on anodes after MOD_{cyc} and OEM_{cyc} protocols. b) Visual juxtaposition of two anodes that have underwent 115 cycles of the MOD_{cyc} protocol (top) and the OEM_{cyc} protocol (bottom) showing the difference in the amount of dead lithium deposits. Marked red boxes refer to terminal tab imprints. c) SEM-EDS analysis of anodes harvested after MOD_{cyc} (top four panels) and OEM_{cyc} protocols (bottom four panels). While a morphologically flat coverage with clear boundaries of lithium deposits is seen on MOD_{cyc} anodes, a more localized concentration of lithium depositions with less pronounced boundaries is observed for OEM_{cyc} anodes.

for all kinds of macroscopic morphologies including lithium plating.[39] For the sake of clarity, it is also important to mention the subsequent reaction pathways after lithium deposition on the anode: i) Deposited lithium can react with electrolyte to form more SEI and other deposits,[39] ii) deposited lithium can electronically insulate from the bulk anode resulting in "dead lithium,"[39,40] iii) dead lithium can react with electrolyte to form more SEI and other deposits,[39] iv) deposited lithium can be intercalated into the anode (during charge),[41] and v) deposited lithium can be stripped away (during discharge).

A first look reveals that in cells cycled with the MOD_{cyc} protocol, anodes exhibit a number of small localized deposits (top panel in Figure 2b), while anodes harvested after OEM_{cyc} show very large deposits covering a much larger area (bottom panel in Figure 2b), similar to those reported previously in the literature.[36] Since all anodes were harvested from fully discharged cells (slow discharge, all cyclable lithium in the cathode), it is reasonable to assume that the observed deposits are either nonreversible lithium deposits that are lost from the inventory of cyclable lithium or reaction products of the highly reducing metallic lithium deposited on the anode reacting with electrolyte components to form SEI.[38,42] A closer look at the harvested anodes reveals that, in cells cycled with the MOD_{cyc} protocol, the anode only shows localized deposits in the folds of the rolled pouch cell, the outer edges (margins), and on the locations of the current terminals. A more homogeneous distribution of deposits can be seen in cells cycled with the OEM_{cyc} protocol. We hypothesize that the terminal depositions are the result of locally higher current density and/or pressure. These features are also observed while opening the cell in the glove box (see Figure S4, Supporting Information), before removing the separator facing the anode; careful comparison shows that some deposits stick to the separator and that they are especially visible at the location of the terminals, edges, and folds. These observations suggest that favorable lithium deposition spots are most likely the edges and folds of the anode roll, as has been shown in previous studies.[43–45]

To explain the variation in deposits between cells we have to look at the main difference between the OEM_{cyc} and MOD_{cyc} charging protocols (Figure 1). While the multistep CC-CV profile of the OEM_{cyc} protocol applies different fixed currents in different fixed voltage regions over the entire lifetime of the battery, the MOD_{cyc} protocol is specifically designed to adapt the applied current as well as the SOC regions in which these currents are applied as a function of battery SOH. The difference between both protocols is visible in Figure 1: higher current rates at higher SOC regions are applied by the OEM_{cyc} while the MOD_{cyc} protocol derates the currents at higher SOC regions. Although the application of higher currents at lower SOCs with subsequent derating, sometimes referred to as

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"boostcharging,"[46,47] is often observed for various LIB charging protocols, in the presented case, it leads to a significantly higher amount of deposits on the anode for the OEM_{cyc} protocol. This is because the anode is more prone to lithium deposition when high current rates are applied specifically in high SOC regions,[42,46] as is for the OEM_{cyc} protocol, and the derating is fixed and not dynamically adjusted with the battery SOH. Nonetheless, deposited lithium does not automatically lead to lost cyclable lithium, as a part of the lithium deposits can reinsert into the graphite anode due to the existing potential difference between deposited (plated) lithium and Li_xC, driving lithium back into graphite.[38] This so-called "relaxation effect" has been studied for overcharged batteries by Gotoh et al.[48] In a similar manner, deposited (plated) lithium that has not been driven back into the graphite anode, can be stripped away during discharge, since the process of lithium plating per se is reversible.[36,49] The mentioned process of lithium reinsertion becomes especially effective for long CV phases.[42] Taking a look at the CV length, one can see a difference in the length of the CV step, especially towards EoL (bottom panels in Figure 1a,b). While the CV step for the OEM_{cyc} protocol lengthens with increased cycle number, most likely due to impedance rise, the CV step for the MOD_{cyc} algorithm is reached at different derated currents and SOCs and stays relatively unchanged over the lifespan of the battery (also see charging times in Figure 1d). Interestingly, a longer CV hold (as for the OEM_{cyc} protocol) would mean more time for lithium reinsertion as discussed by Petzl et al.[42] However, the elongated time of the CV step for the OEM_{cyc} protocols does not seem to trigger a reinsertion of lithium into the anode, which could be due to a prevailing effect of the initially described higher applied currents in high SOC regions leading to more lithium than can be reinserted or due to longer times at higher voltages. Spending more time at higher voltages can cause unwanted oxidation of the electrolyte at the cathode, and therefore transition metal reduction, dissolution from the cathode, and migration to the anode, which then can catalyze thick SEI formation. Since it is beyond the scope of this work to investigate high voltage-related aging mechanisms for the cells under investigation, this topic will be covered in a follow-up study.[50]

Micromorphological investigation of anodes after exposure to air using scanning electron microscopy (SEM) and energy dispersive spectroscopy (EDS) of the depositions reveal that anodes harvested from cells cycled with the MOD_{cyc} protocol show a clear boundary between areas affected by and areas free of lithium deposits. In contrast, the boundary regions of deposits on cells cycled with the OEM_{cyc} protocol are less distinct as seen in the top and bottom panel of Figure 2c, respectively. Areas affected by deposits show high oxygen concentrations in the EDS mapping (orange), while areas not affected by deposits have a clear distinct carbon K_α signal from the graphite anode (red), which is especially visible in the oxygen and carbon specific EDS images. The strong oxygen signal is an indication that deposited lithium as well as lithium components of the SEI are converted to LiOH and Li_2CO_3 (also shown by X-ray diffraction (XRD) and Raman spectroscopy in Figure S5, Supporting Information). Careful timing of the exposure to ambient conditions furthermore reveals that first LiOH is formed and the formed LiOH subsequently transforms into Li_2CO_3 (Figure S5, Supporting Information). Furthermore, areas of lithium deposition on the MOD_{cyc} anode show a very dense and morphologically flat coverage of the electrode surface, while the morphology of lithium depositions on the OEM_{cyc} anode are very localized, and less homogenous (on a microscopic scale), which has been variously associated in the literature with dendritic, needle-like, or "hilly" lithium.[51–54] Longer times at higher currents cause more of the high, needle-like deposits. In contrast, lithium deposits on the MOD_{cyc} anode are widespread and flat. The observed variation in morphology could contribute to greater cell swelling for cells that are cycled with the OEM_{cyc} protocol (Figure S3, Supporting Information), since the dendritic lithium depositions during OEM_{cyc} seem to grow vertically. Lithium deposits on the OEM_{cyc} anode also seem to have a greater surface area, which can lead to more side reactions of dendritic lithium with electrolyte and hence a higher capacity loss.[55,56] Micromorphological characterization of lithium nucleation and growth under various experimental parameters has been thoroughly investigated in the literature.[38,57–61] It has been found that the morphology as well as dendritic growth behavior of deposited lithium is strongly depending on the anion species in the electrolyte,[62] type of carbon used for the anode,[63] external factors such as temperature[51] and also the charging strategy.[52] Being in good agreement with the literature, the presented results illustrate how the charging strategy, especially fast-charging strategy, can influence not only the amount but also the shape of lithium depositions on the anode.

5. Cycling Protocol-Induced Differences on the Cathode

Visually darker spots were observed on cathodes that mirrored the supposed lithium deposits on the anode (highlighted in red in **Figure 3**a–c) for several sets of cells after both OEM_{cyc} and MOD_{cyc} protocols. The smaller amount of lithium deposits formed on anodes subjected to MOD_{cyc} (top panel in Figure 2b), resulted in significantly less cathode darkening (see Figure S6, Supporting Information) overall, but even here this "mirroring effect" was observed. Using ImageJ v1.52a,[64] a quantification of the darker spots revealed that on average 5.7% of the total cathode area is affected by this darkening for MOD_{cyc} cathodes and 17.9% for OEM_{cyc} cathodes. Close-up photographs during careful unwinding of the electrode roll (see Figure S7, Supporting Information) show in detail how exactly deposits on the anode are "mirrored" on the darker cathode regions and that there are no deposits on the separator facing the cathode. Possible reasons for the cathode darkening could be, i) cathode electrolyte interphase (CEI) surface layers and other degradation products remaining on the cathode, ii) morphological changes to the electrode microstructure, iii) chemical composition changes within the cathode material, and iv) a lower lithiation state. However, washing cathodes with diethyl carbonate (DEC) did not change the darker coloring, and hence precludes deposits as a result of unwanted side reactions, e.g., electrolyte oxidation.

SEM-EDS images of the darker areas on the cathode revealed no morphological or chemical differences compared to the lighter regions (Figure S8, Supporting Information). Figure 3d

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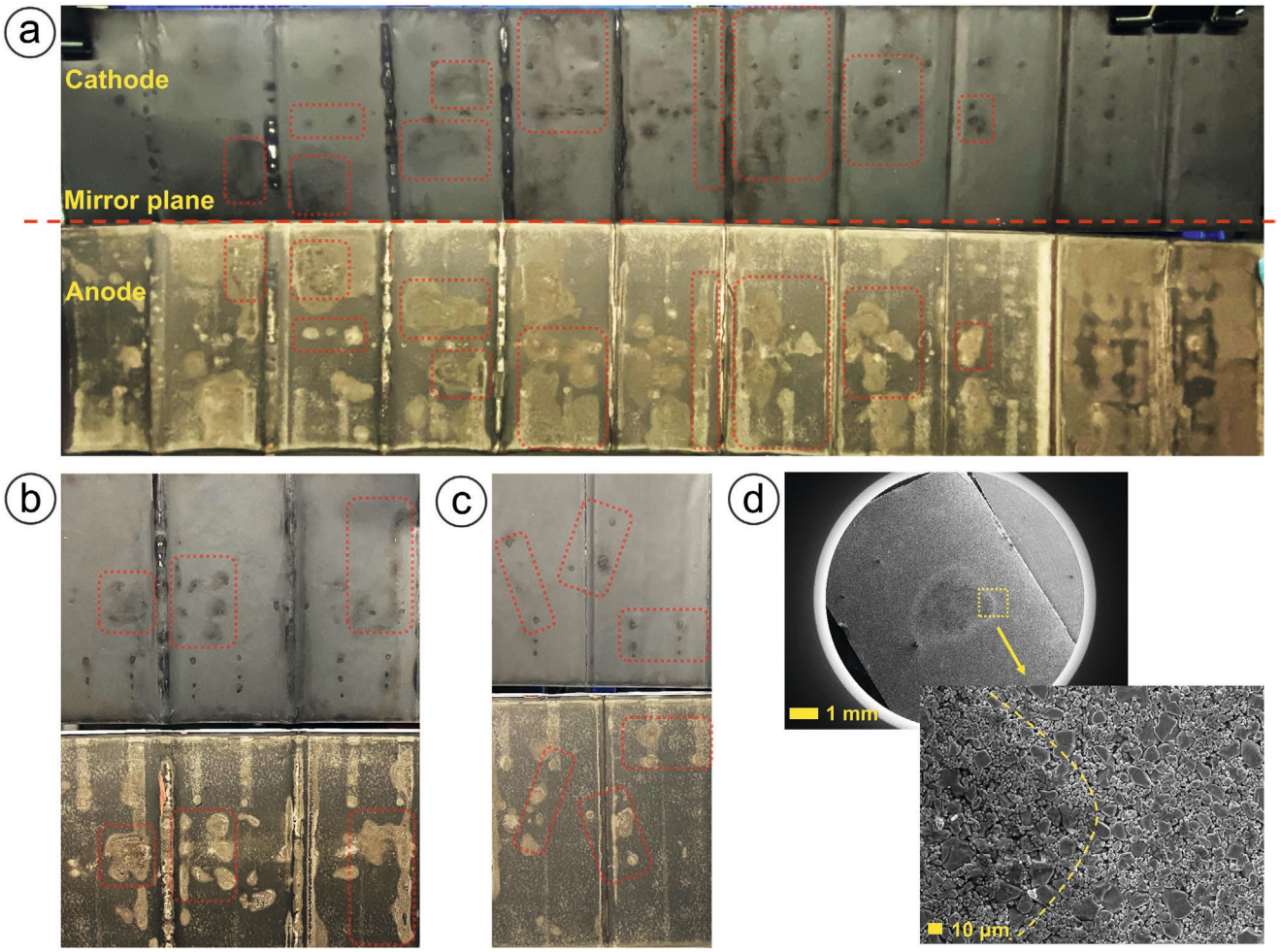


Figure 3. a) Visual juxtaposition of anode and cathode after OEM_{cyc} protocol showing how depositions on the anode are "mirrored" on the cathode. b,c) Magnified images of a second cell confirming that the darkening of the cathode is opposite the lithium deposition sites on the anode. d) Magnified image of a "mirrored" darker spot on the cathode that is opposite the lithium deposition sites on the anode (top), with a higher magnified image of the highlighted area showing the darker regions on the cathode with adjacent cathode areas that are not opposite lithium depositions on the anode (bottom).

is an example of a darker area on the OEM_{cyc} cathode, which is exactly opposite a lithium deposit on the anode. A clear boundary between the visually darker areas and the adjacent areas that do not show darkening is observed and confirmed at higher magnifications. A detailed look at the cathode particle shape and structure confirms the unchanged morphology of cathode particles opposite from lithium deposits showing darkening, which is also confirmed by a detailed analysis of another set of cathodes (Figure S9, Supporting Information). EDS shows that the chemical composition and elemental distribution do not differ between dark regions and adjacent brighter regions. Although it is known that lithium plating can lead not only to deposits on the anode, but also pierce through the porous separator,[54,65] the exact effect of lithium plating on the opposing cathode is not well characterized and little literature is available. Studies that have investigated the morphology of nickel manganese cobalt oxide (NMC) and lithium iron phosphate (LFP) electrodes after aging-induced lithium plating on the anode have found, that the opposing cathode showed no morphological changes,[46,66] visible morphological deterioration,[67] or any distinct change in the active material microstructure after 3000 cycles.[55] One of these studies analyzed aged graphite/NMC cells in which the graphite anode experienced some lithium plating but the NMC cathode showed no major changes in particle structure, crystal structure or color.[66] Inductively coupled plasma optical emission spectroscopy analysis of the cathode material, however, showed that it contained a reduced amount of lithium.[66] Since anodes after OEM_{cyc} showed large amounts of lithium deposits, a lower lithium inventory is expected and hence could result in a lower lithium content on opposing darker areas on the cathode. To explore this hypothesis, we thoroughly investigated the observed darker cathode regions in the following.

Figure 4a shows the XRD diffraction patterns that are obtained from cathodes unaffected by darkening and cathodes that show severe darkening on the surface. A clear shift in the diffraction peaks between both regions can be seen, indicating a difference in the LCO structure. A search against a Powder Diffraction File (PDF) database confirms the initial hypothesis of a lower degree of lithiation within the LCO structure and results in the best match being $Li_{0.51}CoO_2$ (PDF 01-083-6088).

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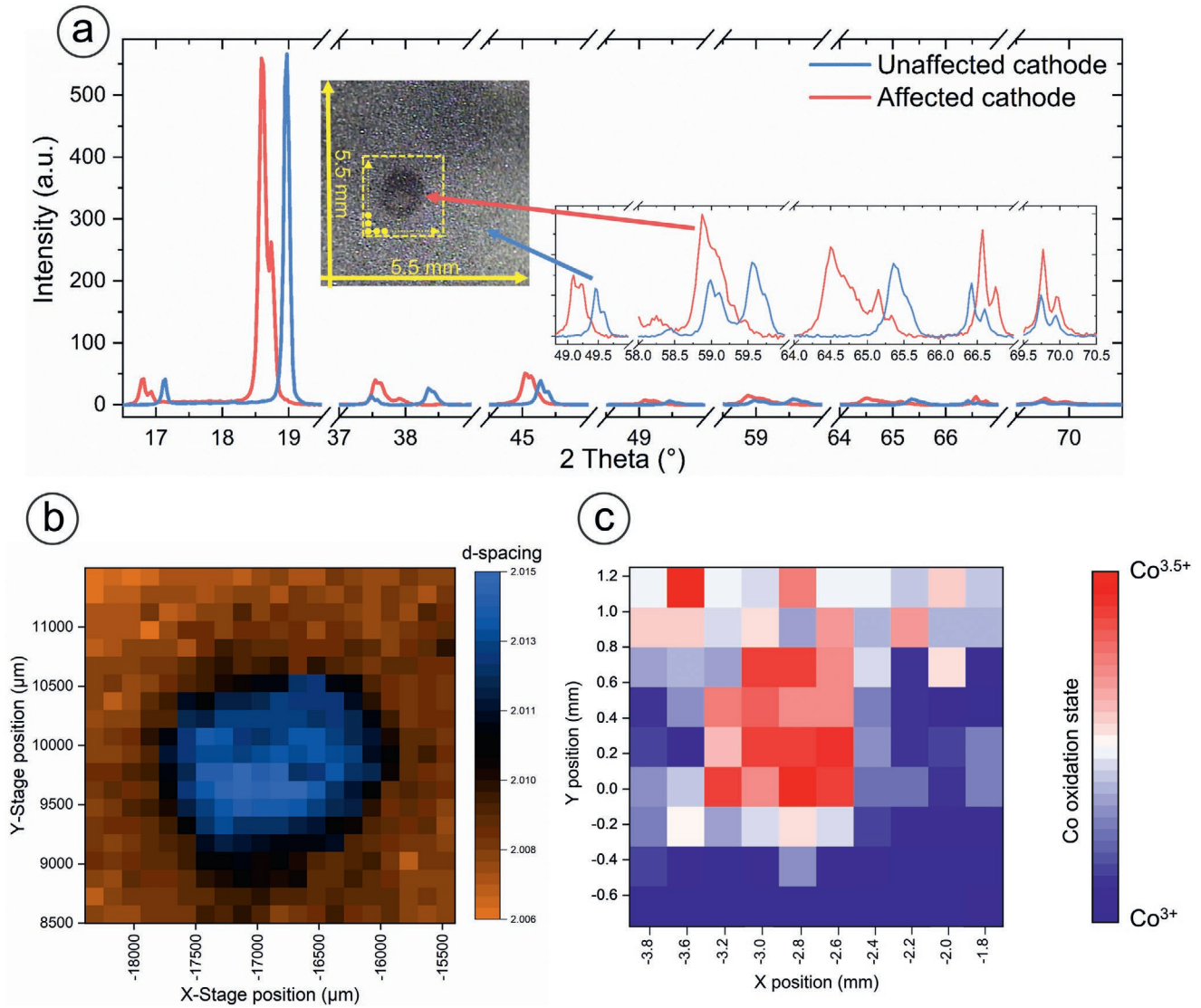


Figure 4. a) XRD of harvested cathodes after MOD_{cyc} (blue) showing no significant darkening and cathodes after OEM_{cyc} (red) showing significant darkening; a great difference in lithiation was observed when comparing the obtained XRD spectra to LCO with various lithium contents. b) Synchrotron micro-XRD analysis of a darkened cathode spot reveals a clear difference in the d-spacing extracted from the (104) and (111) reflections of the rhombohedral and monoclinic phases. c) Micro-X-ray absorption spectroscopy (micro-XAS) analysis of the darkened spot shown in the inset of panel (a). Linear combination fitting of the area to a fully delithiated cathode harvested from a pristine cell after full charge shows that the darkened spots are similar to more delithiated cathode, i.e., where cobalt is in a higher oxidation state.

Rietveld refinement for cathodes that are not opposite lithium deposition sites (hereafter referred to as "unaffected cathode"), and darker cathode sites that are opposite anode sites that show severe lithium deposits (hereafter referred to as "affected cathode") reveal structural differences. The phase of the unaffected cathode areas is identified to be a rhombohedral ($R\bar{3}m$) phase, whereas the affected cathode areas appear to be a monoclinic (C2/m) phase (see Figure S10, Supporting Information). Lattice parameters of the rhombohedral phase are determined to be a = 2.8135(5) Å, c = 14.0890(2) Å; lattice parameters of the monoclinic phase are determined to a = 4.8619(7) Å, b = 2.8093(1) Å, and c = 5.0455(7) Å, and β = 108.71(6)°. The Rietveld refinement seems to be in good agreement with what has been previously published about the lattice parameters and the phase transformation of LCO upon delithiation from rhombohedral to monoclinic[68–73] and proves the initial hypothesis of a lower lithiation grade in the affected cathode areas. This result is in very good agreement with a recent study by Wang et al. indicating how the loss of cyclable lithium is resulting in lithium vacancies within the LCO cathode material.[87] To further resolve this phase transition on a micrometer scale we have adopted micro-X-ray diffraction (micro-XRD) and micro-XAS as seen in Figure 4b,c, respectively. Figure 4b shows the change in d-spacing of the (104) and (111) reflections of the rhombohedral and monoclinic phase, respectively. While unaffected areas around the cathode darkening show a smaller d-spacing, the affected areas show higher d-spacing values due to a transition to the monoclinic phase. Repeated phase transitions from the

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rhombohedral O3-1/H1 phase to the O3-2/H2 phase and the monoclinic M1 phase has previously been regarded as a limiting factor for longevity in commercial LCO batteries,[4] however, was shown not to impact the electrochemical performance of LCO negatively.[4,74,75] On the other hand, above 4.5 V, additional phase transitions from the M1 phase to an H1-3 hybrid structure of the rhombohedral form of LCO and CoO_2 and finally an O1 phase, which is made up of only CoO_2 layers in an ABAB stacked form, has been shown to lead to structural degradation and performance loss. This irreversible phase transition to the H1-3 and O1 phase at > 4.5 V exhibits strain on the lattice structure, leads to capacity fade, and oxygen evolution.[4,76–79] Since both MOD_{cyc} and OEM_{cyc} protocols use an upper cell voltage cutoff of 4.4 V, we do not expect a structural degradation of the LCO crystal lattice to the detrimental H1-3 or even O1 phase. Nevertheless, it cannot be excluded that there are no structural changes within the crystal lattice, especially at the surface, since it has been shown previously, how the surface structure and composition of LCO can alter with cycling.[4] For example, Wang et al. showed how the surface structure of cycled LCO can show fracturing, defects, and cation disorder due to exchange of lithium and cobalt from corresponding layers and spinel disordering.[80,81] To what extent this applies to the affected cathode regions in our cells remains unclear, since the performed XRD measurements are a bulk technique.

The observed change in crystal structure for darker cathode areas is also reflected in the oxidation state of cobalt. Figure 4c shows a darkened spot after OEM_{cyc} and the respective linear combination fitting to a fully delithiated cathode (fully charged cell, cobalt oxidation state = 3.5+) collected from a pristine cell after a single full charge. The fit indicates that darkened spots on cathodes harvested after 115 cycles of OEM_{cyc} in a fully discharged state are more similar to fully delithiated cathodes, i.e., where cobalt is in an oxidation state of 3.5+ ($x = 0.5$), and hence seem not to be able to intercalate as much lithium as the unaffected surrounding cathode (also see Figure S11, Supporting Information). The observed change in oxidation state was further analyzed by probing bulk electrodes with hard X-ray absorption spectroscopy (hard XAS), i.e., by collecting cobalt K-edge XAS spectra continuously during cycling of in operando coin cells. Details on the experimental procedure and the design of the multicoin cell holder for the in operando experiment can be found in the Experimental Section and Figure S12 in the Supporting Information. It is important to note here that cathodes most affected by cathode darkening were chosen and that the darker regions were aligned with the beam. **Figure 5**a,b show how harvested cathode materials, after OEM_{cyc} and MOD_{cyc} protocols, have almost identical (de-)intercalation activity when being cycled in an in operando cell with a *C*/10 rate. The energy values shown in Figure 5a were extracted using the edge position defined by the photoelectron energy origin (E_o)[82] and show an almost perfect overlap for cathodes harvested after both cycling protocols, and even show that both cathodes are fully charged after the same time. However, the same energy values for the OEM_{cyc} cathode at the end of the discharge (at 3 V) indicate that the discharge for this cell is completed slightly (≈2 h) earlier than the discharge for the MOD_{cyc} cathode. This can be confirmed by the electrochemical charge/discharge profiles in Figure S13a in the Supporting Information. Similar to the edge energy positions, the cobalt K-edge spectrum for the electrodes in a fully discharged (blue) and a fully charged (red) state displayed in Figure 5b shows a perfect overlap of the cobalt K-edge between the MOD_{cyc} (solid) and the OEM_{cyc} (dotted) absorption edge. In contrast to this, in operando cycling of an unaffected cathode after MOD_{cyc} and an affected cathode after OEM_{cyc} at a *C*/5 rate show a striking difference in the (de-)intercalation cobalt oxidation state change as seen in Figure 5c,d. The minimal redox activity observed for the affected cathode region after OEM_{cyc} is visible in the extracted edge energies and also the overlap of the cobalt K-edge absorption edge, Figure 5c,d, respectively. The cobalt K-edge barely changes for the affected OEM_{cyc} cathode, whereas the unaffected MOD_{cyc} cathode shows considerable activity. Furthermore, in the fully discharged state, cathodes after OEM_{cyc} show higher oxidation states (blue dots, greater shift towards higher edge energies) than MOD_{cyc} cathodes (blue line), indicating that the overall average cobalt oxidation state in the bulk cathode material which underwent OEM_{cyc} is higher, i.e., has a lower lithium inventory. Figure S13b in the Supporting Information shows that the obtained capacity in the in operando coin cell containing the affected cathode is only half of the expected capacity, which is due to the observed inactivity of the darker cathode regions as confirmed in Figure 5c. Only cathode regions that do not show this darkening contribute to the total capacity of the cell. Additional rate tests (Figure 5e) of lithium half-cells at higher rates as used during in operando cycling confirm that the rate performances of the two types of cathodes are significantly different at higher rates. At the highest chosen rate of C/2, half-cells with unaffected cathodes after MOD_{cyc} deliver a discharge capacity of 4.4 mAh, while half-cells with affected cathodes after OEM_{cyc} only deliver 1.3 mAh, three times less, at *C*/3 the difference between obtained discharge capacities is roughly two times. The observed activity of the cathode at very slow rates, when excess lithium is available, confirms that the LCO is still electroactive. In the full cells, however, the affected cathode regions cannot participate in the intercalation of lithium, because of the loss of lithium to mossy lithium deposits. Some of the factors defining rate performance in LIBs are ion migration and electron transfer between the interfaces of electrodes and electrolytes. Excellent ionic and electronic conductivities of electrodes are needed especially for fast-charging applications.[83] Here, the electrolyte/cathode interface accounts at least partially for the poor rate capability of the full cells; in addition, the unwanted reaction products on the anode will possible also increase the cell resistance. It has previously been shown that high voltage cycling can lead to side reactions with electrolyte, increased CEI film thickness, and hence higher impedance of the positive electrode.[84–86] Although the harvested cathodes are washed certain oxidation products may still remain on the surface. Similarly, changes in the surface structure, which are not detectable by the methods described so far, could explain the poor capacity retention upon higher rates. A change in the surface of the cathode material could cause low ionic conductivities at the interface of electrolyte and cathode and would hinder the transfer of lithium ions into the cathode at higher rates. Recently, Wang et al. have reported how a degraded surface layer in LCO particles results in an increased charge transfer resistance because of a

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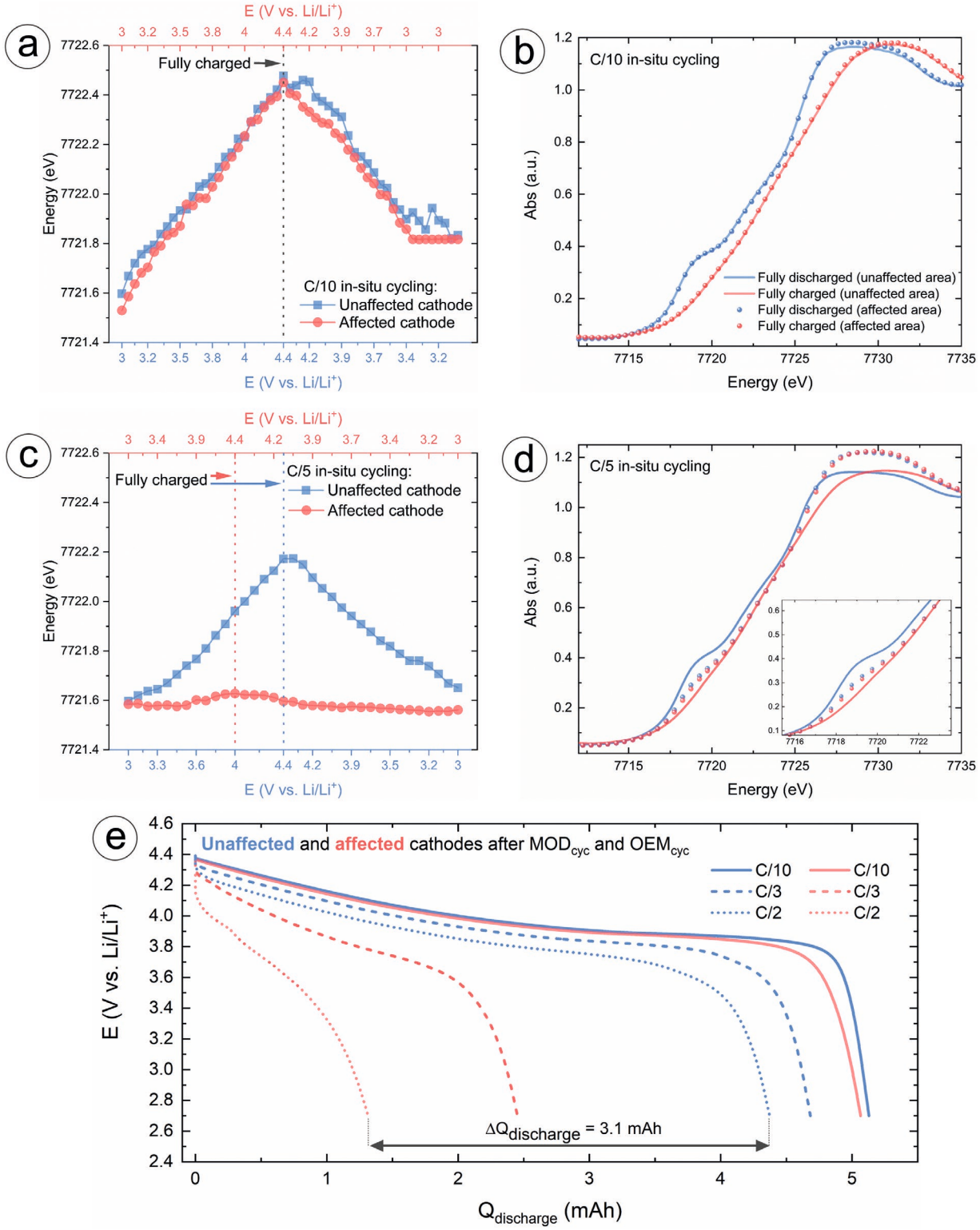


Figure 5. a) Cobalt K-edge energies collected during in operando cycling (C/10) of cathodes affected and unaffected by darkening, harvested after MOD_{cyc} (blue) and OEM_{cyc} (red) protocols. b) Cobalt absorption edges for fully discharged (blue) and fully charged (red) state for MOD_{cyc} (solid) and OEM_{cyc} (dotted) cathodes. c) Cobalt K-edge energies collected during in operando cycling (C/5) of an MOD_{cyc} cathode (blue) not showing any darkening and OEM_{cyc} cathode (red) with significant darkening. d) Cobalt absorption edges at a fully discharged (blue) and a fully charged (red) state for MOD_{cyc} and OEM_{cyc} cathodes. e) Rate capability test for half-cells with metallic lithium anodes for unaffected and affected cathodes harvested after both protocols (for the sake of comparison, only the last profiles of five consecutive cycles at the same rate for each cell are shown). At C/10, unaffected and affected cathodes differ very little, while a $\Delta Q_{discharge}$ of 3.1 mAh is observed at C/2 rate, indicating that the rate capabilities of cathodes showing darkening were strongly affected.

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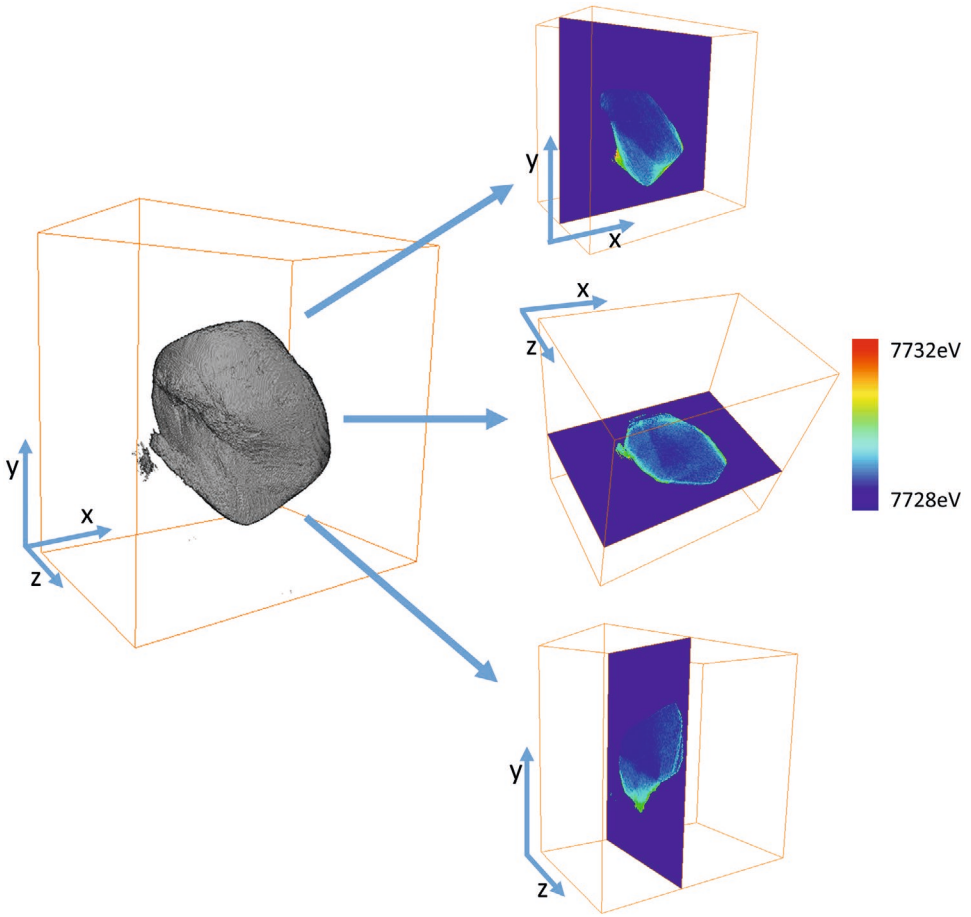


Figure 6. Single particle TXM analysis of an LCO particle harvested from a darkened cathode region after OEM_{cyc} at fully discharged state. Left: 3D rendering of the X-ray nanotomographic data of the particle. Right: cobalt K-edge white-line peak energy over different virtual slices through the center of the same particle shown on the left.

deterioration of the lattice structure stability.[87] Note that surface layers would likely be only several nm thin, so that they would not be detected in the SEM-EDS analysis (Figure 3d and Figures S8 and S9, Supporting Information).

Lastly, we used a single particle characterization technique, synchrotron transmission X-ray microscopy (TXM) to visualize the oxidation state distributions of cobalt within a single LCO particle that was harvested from a darkened region on the cathode electrode after OEM_{cyc} at the fully lithiated state. As shown by our previous characterization techniques, at the end of discharge, the darkened LCO material in the darkened region generally exhibits a lower lithiation degree and higher cobalt oxidation state. Zooming in on a single particle, our analysis with TXM reveals the reaction heterogeneity at the sub-particle level, featuring a more oxidized state of cobalt in the near-surface region in comparison to the particle core, which reached to a more lithiated state with lower cobalt valence. Such depth dependent cobalt redox profile is clearly revealed by the cobalt white-line peak energy distribution over the three virtual slices through the particle center shown in **Figure 6**.[88] This observation is in good agreement with the particle-level study of the other cathode materials and clearly indicates that the surface,[89] more than the bulk, is the bottleneck for the lithium intercalation at higher charge/discharge rates as also observed in the rate test, i.e., higher oxidations states, for the lower lithiation grades in the darker cathode regions are not homogenously distributed over the entirety of a single particle. It is also important to note here, that the presented TXM result of a single particle might not be representative of the results obtained from bulk techniques, where several thousands of particles are probed simultaneously. A more detailed TXM study focusing on the particle-level related degradation of LCO is necessary to understand the obtained results and get statistically representative data.

In closing, we would like to offer possible hypotheses as to how the darkened cathode spots can remain at a lower lithiation state and concomitant higher cobalt oxidation state:

1) Continuous lithiation of the anode upon fast-charging requires fast lithium diffusion in order to prevent lithium concentration gradients. The formation of dead lithium on the anode locations opposite of the darkened cathode locations hinders fast lithium diffusion in and out of the anode. The lateral lithium diffusion into the ≈1–2 mm diameter darkened cathode regions takes on the order of 10–20 h (assuming an electrode tortuosity of 3–5, and a diffusion

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coefficient of 3.5×10^{-6} cm^2 s^{-1}, value from Landesfeind and Gasteiger[90,91]), which means that the darkened regions opposite of anode deposits cannot be (de-)lithiated under the fast charge/discharge cycling (see Figure 1). However, this does not explain why darkened cathode regions stay in a lower lithiation state at slow discharge (C/20).

2) The observed lower lithiation states within darkened cathode regions could be related to surface degradation of cathode particles. Degradation of the LCO particle surface can manifest as fracturing, lattice defects, and cation disorder as well as spinel layer formation.[80,81] This in turn can influence the material's rate capability for lithium (de-)intercalation. As discussed in the XRD section and shown with half-cells as well as in operando hard XAS measurements (see Figure 5), slow rate cycling in lithium metal half-cells can reactivate darkened cathode regions. However, in the aged full cells even slow discharging cannot relithiate the darkened cathode regions such that surface degradation by itself is insufficient to explain the observed phenomenon.
3) The reason why darkened cathode regions in full cells cannot be relithiated even at slow discharge rates, can be linked to a possible loss of cyclable lithium and electrolyte. The formation of thick layers of decomposition products on the anode result from the consumption of electrolyte and a subsequent electrolyte "dry-out." Typically, no excess of electrolyte is used in commercial smartphone batteries, which could result in local electrolyte starvation, and hence the darkened cathode spots with lower lithiation state. Further, the aggregated layer thicknesses of deposits on the anode (see Figure 2b and Figure S4, Supporting Information), with increasing cycle number, will likely cause local compression on separator and cathode in the tightly packed hard-case smartphone cell (compare to our cell swelling data in Figure S3, Supporting Information). This can lead to electrolyte being pressed out and also result in local electrolyte starvation. We think this could be a likely hypothesis for the fact that even under slow discharge the darkened cathode regions opposite of lithium deposits cannot be lithiated.

6. Conclusion

In this work, we highlighted the critical importance of a computationally inexpensive, electrochemical model-based fast-charging algorithm on capacity fade for LCO battery chemistry. Higher capacity retention, less swelling, and almost no lithium deposition were the result of a highly tailored fast-charging algorithm adapting voltage and current limits according to the SOC/SOH of the battery. Significantly higher amounts of lithium deposits and more dendritic lithium were observed on the anode for cells undergoing OEM charging compared to the model-based charging. The main cause of capacity fade, the loss of cyclable lithium, is also shown to result in inactivated cathode areas directly opposite to or "mirrored" to thick lithium depositions on the anode. A thorough analysis showed that the affected cathode regions do not participate in (de-) intercalation and are in a permanently lower lithiation state, exhibit a monoclinic phase, and contain cobalt in a higher oxidation state, although they did not show any morphological or chemical difference compared to unaffected areas. The intercalation activity of affected cathode regions could be nearly completely recovered at slow rates when excess lithium in form of a lithium metal anode is employed. At higher rates, affected cathode regions did not show any activity, and harvested cathodes with many affected regions delivered low reversible capacity. The results suggest that the inactivity at higher rates is a result of surface degradation of cathode particles opposite from lithium deposits on the anode.

7. Experimental Section

Battery Specifications and Cycling: The LG-G5 cell has a nominal capacity of 2.8 Ah, and a maximum cell voltage of 4.4 V, and the cell temperature should be kept below 40 °C. Two charging protocols were used for the LG-G5 battery. The standard charge is a CC-CV (constant current – constant voltage) protocol with a constant current at 1.35 A and a CV hold at 4.4 V. The fast charge protocol (referred to as OEM_{cyc} charge in this paper) is a multistep CC-CV protocol with the first step at 3.105 A until 4.30 V, the second step at 1.89 A step until 4.4 V followed by a CV hold at 4.4 V. Since the main focus of this paper is the impact of fast-charging protocols on the battery aging behavior, the multistep CC-CV fast charge protocol was studied in more detail.

In order to make a fair comparison between different charging protocols, the OEM fast-charging algorithm (OEM_{cyc}) was compared with the model-based fast-charging algorithm (MOD_{cyc}) with similar 0–100% SOC charge times. The model-based charging protocols rely on minimizing certain aging indicators through the charging stage while simultaneously minimizing the time required to fully charge the battery, i.e., as the cell ages, the optimal fast-charging profile is approximated using simple building blocks for profile generation and adaptation rules to match the aged state of the cell.

The testing protocol consists of performance tests at BoL, after every 50 cycles and EoL. EoL is defined as the point where the capacity during cycling reaches 80% of the capacity at BoL. The performance test comprises of a C/25 rate slow charge and discharge between 2.7 and 4.4 V to characterize the true capacity of the battery, the analysis of $dQ - dV$ trends as a function of battery age, and swelling measurements using a digital caliper. The cycling test was comprised of a charge (OEM_{cyc} or MOD_{cyc}) step (see above), rest after charge step (for 30 min), followed by discharge (drive cycle profile, constant current discharge at multiple C-rates) until 2.7 V and finally a rest after discharge step (for 30 min). This sequence of steps was then repeated until EoL. In order to facilitate a fair comparison in the postmortem analysis, the cycling was stopped once the cells reached the EoL capacity criterion. The cells were studied using analytical techniques at the same cycle numbers. Since the discharge profiles for the cells remained the same and the cells were charged with the same CV cutoff criterion, the difference in aging between the OEM_{cyc} and MOD_{cyc} batteries is attributed to the charging strategy.

All tests were performed using a Arbin BT2000 tester and in a controlled temperature environment at 25 °C. The battery tester has current/voltage accuracy of 0.02%. In order to access the cell terminals the battery Protection Circuit Boards (PCBs) were removed.

X-Ray Diffraction Analysis: Lab scale X-ray diffraction patterns of harvested anodes and cathodes were recorded using a Bruker D8 ADVANCE X-ray diffractometer equipped with a Cu K_α radiation source and a Bruker LYNXEYE XE detector. The accelerating voltage and current were 40 kV and 40 mA, respectively. Scans were collected between 10° and 80° (2θ) at step size of 0.03° and a 1 s time per step. All samples were rotated at a 10° min^{-1} rotation speed. Background correction for all gathered XRD spectra was done using the Bruker DIFFRAC.EVA v4.2.1.1 software suite. A PDF database version x was used. For the refinement, GSAS-II v.3957 was used.[92]

Raman Spectroscopic Analysis: Raman spectroscopy was performed using a Horiba Scientific LabRAM HR Evolution Raman microscope equipped with a 405 nm diode laser, an 1800 grooves per mm grating, and a Synapse Open-Electrode charge-coupled device detector. Laser

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focusing was done using an Olympus lens with a 50 x magnification, a numerical aperture of 0.5, and a field number of 26.5. An area of ≈50×50 μm^2 was probed by the laser. Scattered radiation was collected in a backscattering geometry through the same lens. Rayleigh-scattered laser light was removed using a filter prior hitting the spectrometer. Several spots were measured and averaged to assure representative spectra collection, spectra were collected between 50 and 1500 cm^{-1}.

Scanning Electron Microscopy-Energy Dispersive Spectroscopy: Scanning electron microscopy imaging and energy dispersive X-ray spectroscopy were conducted using a JEOL JSM-7200F Field Emission Scanning Electron Microscopy equipped with an Oxford X-Max^N 50 Silicon Drift Detector system. SEM imaging was performed using an accelerating voltage of 5 kV at medium probe currents using the secondary electron detector. EDS was conducted using an accelerating voltage of 20 kV at high probe currents.

Hard X-Ray Absorption Spectroscopy: Hard XAS ex situ and in operando data on cobalt K-edge was collected in transmission mode using a Si (220) monochromator at SSRL beamline 2-2 and 4-1, respectively. For ex situ analysis, harvested cathodes were carefully transferred on Kapton films after washing. In operando cells were built as described in the separate section below. All sample handling was conducted in an argon filled glovebox (O_2 < 0.1 ppm, H_2O < 0.1 ppm). Higher harmonics in the X-ray beam were rejected by detuning the Si (220) monochromator by 40% at the nickel K-edge. Energy calibration was accomplished by using the first inflection points in the spectra of cobalt metal foil reference at 7709 eV. XANES data were analyzed by Sam's Interface for XAS Package (SIXPACK),[93] with the photoelectron energy origin (E_0) determined by the first inflection point of the absorption edge jump. Prior to data acquisition, samples were mounted into a sample box with flowing inert gas.

In Operando Coin Cell Preparation and Measurements: An overview of the in operando coin cell holder and cell preparation can be seen in Figure S12 in the Supporting Information. 2 mm diameter holes were drilled into stainless steel Hohsen 2032 coin cell bottom and top caps using a drill press. A 5 mm hole was drilled into Hohsen stainless steel spacers of 1 mm thickness and 15 mm diameter using the same drill press. Sharp edges were filed off with a Dremel rotary tool using the silicon carbide grinding stone. Similarly, the surface around the holes in the top and bottom coin cell caps was roughened for later application of epoxy. Kapton tape discs of 8 mm diameter and 0.127 mm thickness were punched out using an arch punch. All coin cell parts, spacers and the Kapton discs were sonicated with ethanol for 30 min and dried overnight at 60 °C. Kapton discs were glued onto the top and bottom coin cell parts using epoxy. All parts were vacuumed overnight in a glove box antechamber prior to bringing the items into the glove box. In operando coin cells were built using Rockwood 450 μm thick lithium with 15 mm diameter, two Celgard separators with 18 mm diameter and the harvested cathodes with a 14 mm diameter after OEM_{cyc} and MOD_{cyc} protocols. 30 μL of 1 M $LiPF_6$ in ethylene carbonate (EC): diethyl carbonate (DEC) (50:50, wt%) was used as an electrolyte. Cells were taken out of the glove box and a second layer of epoxy was applied around the Kapton discs that were glued on top of the holes of the bottom and top coin cell caps. Prior to taking the in operando cells to the beam line, a formation step was performed after a 6 h equilibration time using a *C*/10 rate. In operando cells were transported to the beam line in a fully discharged state. A custom three cell holder was built using acrylic glass and single coin cell holders (Keystone Electronics, 1060), where a 5 mm diameter hole was drilled into with a drill press (see Figure S12, Supporting Information). Long copper wires were soldered to each coin cell holder for connecting to a potentiostat.

Cathode Half-Cell Building and Cycling: Half-cells with harvested cathodes for rate capability tests were built in a similar manner as described for the in operando coin cells, however using undrilled coin cell caps and spacers. No changes in the diameters of the used components were made. Half-cells were cycled using a BioLogic VMP3 potentiostat. Repeatability of the rate tests was checked by using five or ten cycles for each C-rate. Chosen *C*-rates were *C*/10, *C*/5, *C*/4, *C*/3, and *C*/2.

Synchrotron Micro-X-Ray Diffraction: Synchrotron micro-XRD measurements were carried out at beam line 12.3.2 at the Advanced Light Source, Lawrence Berkeley National Laboratory. The unique capability of this station is a micrometer focused synchrotron beam allowing acquisition of diffraction spectra on a micrometer scale using a DECTRIS Pilatus 1 M pixel array detector. Harvested cathode samples after OEM_{cyc} were fixed on a metal block and mounted into the hutch. A 3×3 mm^2 area was probed using a 10 keV beam with a step size of 150 μm. Obtained data were processed using the X-ray microdiffraction analysis software suite.[94]

Synchrotron Micro-X-Ray Absorption Spectroscopy: Synchrotron micro-XAS analysis of harvested cathodes affected by darkening after OEM_{cyc} was conducted at beam line 2–3 at the Stanford Synchrotron Radiation Lightsource (SSRL), a division of the Stanford Linear Accelerator (SLAC) National Laboratory. A Kirkpatrick-Baez system was used to focus the beam down to a spot size of 2×2 μm^2. Detection was achieved with a Vortex silicon drift detector. Areas of 1.8×1.8 mm^2 size were probed with X-rays. A step size of 200 μm in *x* and *y* direction was used to acquire 100 spectra within the probed area. Acquired data were analyzed using the SIXPack software suite,[93] with the photoelectron energy origin (E_0) determined by the first inflection point of the absorption edge jump.

Transmission X-Ray Microscopy: TXM measurements of harvested single particles were conducted at beam line 6-2c at Stanford Synchrotron Radiation Lightsource (SSRL), a division of the SLAC National Accelerator Laboratory. More details of this experimental approach can be found in the literature.[95] The TXM data analysis was carried out using an SSRL in-housed developed software package.[96]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The synchrotron experiments of this research were performed at the Stanford Synchrotron Radiation Lightsource (SSRL), a Directorate of SLAC National Accelerator Laboratory and an Office of Science User Facility operated for the U.S. Department of Energy Office of Science by Stanford University. Use of the Stanford Synchrotron Radiation Lightsource, SLAC National Accelerator Laboratory, was supported by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences under Contract DE-AC02-76SF00515. Work at the Molecular Foundry was supported by the Office of Science, Office of Basic Energy Sciences, of the U.S. Department of Energy under Contract DE-AC02-05CH11231. This research used resources of the Advanced Light Source, a U.S. Department of Energy Office of Science User Facility under Contract No. DE-AC02-05CH11231.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

fast-charging, lithium cobalt oxides, lithium-ion, physics-based models, smartphones

Received: September 22, 2020
Revised: January 8, 2021
Published online: February 16, 2021

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